7. Write a Servlet program to store a user's preferences (like theme or language) using cookies. Retrieve and display these preferences on subsequent visits.

AIM

To create a Java Servlet that stores user preferences (theme and language) using cookies and retrieves them on subsequent visits.

ALGORITHM

1. Create a Dynamic Web Project in Eclipse.
2. Configure Apache Tomcat Server.
3. Create an HTML form to collect user preferences.
4. Develop a Servlet to:
5. Store preferences in cookies
6. Retrieve cookies when user revisits
7. Deploy the project on the server.
8. Run the application in a browser.

SOURCE CODE

1. HTML Page (index.html)

```
<!DOCTYPE html>
<html>
<head>
    <title>User Preferences</title>
</head>
<body>
    <h2>Select Your Preferences</h2>
    <form action="PreferenceServlet" method="post">
        Theme:
        <select name="theme">
            <option value="Light">Light</option>
            <option value="Dark">Dark</option>
        </select>
        <br><br>

        Language:
        <select name="language">
            <option value="English">English</option>
            <option value="Tamil">Tamil</option>
        </select>
        <br><br>

        <input type="submit" value="Save Preferences">
    </form>
</body>
```

```
</html>
```

2. PreferenceServlet.java

```
import java.io.; import javax.servlet.; import javax.servlet.http.*;

public class PreferenceServlet extends HttpServlet {

protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

    String theme = request.getParameter("theme");
    String language = request.getParameter("language");

    // Create cookies
    Cookie themeCookie = new Cookie("theme", theme);
    Cookie languageCookie = new Cookie("language", language);

    // Set expiry time (1 day)
    themeCookie.setMaxAge(24 * 60 * 60);
    languageCookie.setMaxAge(24 * 60 * 60);

    // Add cookies to response
    response.addCookie(themeCookie);
    response.addCookie(languageCookie);

    // Redirect to display page
    response.sendRedirect("display");
}


}
```

3. DisplayServlet.java

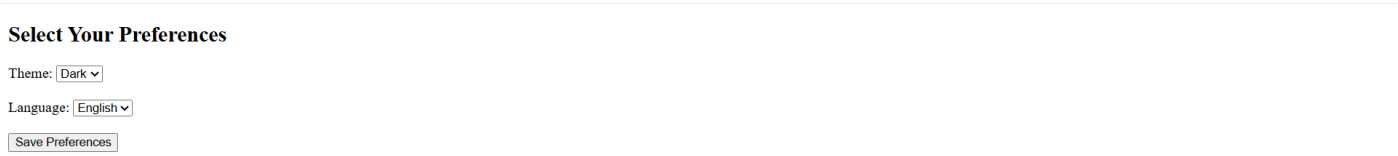
```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class DisplayServlet extends HttpServlet {

   protected void doGet(HttpServletRequest request, HttpServletResponse response)
         throws ServletException, IOException {

      response.setContentType("text/html");
      PrintWriter out = response.getWriter();

      String theme = "Not Set";
      String language = "Not Set";

      Cookie[] cookies = request.getCookies();

      if (cookies != null) {
         for (Cookie c : cookies) {
            if (c.getName().equals("theme")) {
               theme = c.getValue();
            }
            if (c.getName().equals("language")) {
```

```
                language = c.getValue();
            }
        }
    }

    out.println("<html><body>");
    out.println("<h2>User Preferences</h2>");
    out.println("<p>Theme: " + theme + "</p>");
    out.println("<p>Language: " + language + "</p>");
    out.println("<a href='index.html'>Back</a>");
    out.println("</body></html>");
  }
}
```

4. web.xml

```
<web-app>

  <servlet>
    <servlet-name>PreferenceServlet</servlet-name>
    <servlet-class>PreferenceServlet</servlet-class>
  </servlet>

  <servlet-mapping>
    <servlet-name>PreferenceServlet</servlet-name>
    <url-pattern>/PreferenceServlet</url-pattern>
  </servlet-mapping>

  <servlet>
    <servlet-name>DisplayServlet</servlet-name>
    <servlet-class>DisplayServlet</servlet-class>
  </servlet>

  <servlet-mapping>
    <servlet-name>DisplayServlet</servlet-name>
    <url-pattern>/display</url-pattern>
  </servlet-mapping>

</web-app>
```

OUTPUT :



Result:

The Servlet successfully stores user preferences using cookies and retrieves them on subsequent visits.